Enhancing Grade 8 Geometry Learning through Offline-Capable Augmented Reality: A Case Study in Sri Lanka

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***Abstract*—Mathematics education in Sri Lanka, particularly at the Grade 8 level, faces challenges in developing students' spatial understanding because geometry concepts are often taught using static two-dimensional textbook illustrations. Concepts such as solids, quadrilaterals, and triangles require learners to visualize spatial relationships that are difficult to represent through flat images alone. To address this problem, this study developed an offline-capable Augmented Reality (AR) application that transforms selected Grade 8 geometry concepts into interactive three-dimensional models. The application was designed for low-specification mobile devices and includes quiz activities to reinforce learning and encourage active participation. A quantitative evaluation with 59 Grade 8 students examined system engagement and quiz-based academic performance. Students spent between 350 and 600 seconds using the application, and the overall mean quiz score was 2.51 out of 5. The findings indicate that AR supports engagement, but instructional scaffolding is needed to improve learning outcomes.**

***Index Terms*—Augmented Reality, Mathematics Education, Geometry Learning, Interactive Learning, 3D Visualization, Educational Technology**

I. Introduction

Geometry learning requires students to understand shapes, angles, spatial relationships, and three-dimensional structures. For Grade 8 students in Sri Lanka, these concepts are commonly introduced through textbook illustrations and teacher-centered classroom instruction. Although such methods provide basic conceptual explanations, they often limit students' ability to visualize and manipulate geometric objects. This becomes especially challenging when learners are required to understand three-dimensional structures using flat two-dimensional representations [1], [2].

Spatial reasoning is an important cognitive skill in mathematics and is strongly connected to future learning in science, technology, engineering, architecture, and related disciplines. When students are unable to visualize geometric concepts effectively, their ability to solve geometry problems and apply mathematical reasoning may be reduced. In the Sri Lankan classroom context, this issue is further influenced by limited access to advanced digital learning tools, uneven availability of internet connectivity, and the need for curriculum-aligned resources that can function in resource-constrained environments [5], [6]. Augmented Reality (AR) provides an opportunity to address these limitations by placing interactive digital objects within the learner's real-world environment. In geometry education, AR allows students to rotate, zoom, and inspect three-dimensional models from different viewpoints. This can make abstract geometric concepts more concrete and can support active learning. However, many existing AR-based educational tools require continuous internet access, high-end devices, or content that is not aligned with the Sri Lankan Grade 8 mathematics curriculum [6], [7].

To address this gap, this study presents an offline-capable AR application designed specifically for Grade 8 geometry learning in Sri Lanka. The application converts selected textbook-based geometry concepts into interactive three-dimensional models and integrates quizzes to assess student understanding. The main objective of this study is to evaluate the effectiveness of the AR application in terms of student engagement and quiz-based academic performance.

The main contribution of this study is the development and evaluation of an offline-capable AR learning application aligned with Grade 8 geometry topics in Sri Lanka. The application integrates interactive three-dimensional geometry models and quiz-based learning activities into a mobile AR environment. In addition, the study provides a quantitative evaluation of student engagement and academic performance using logged interaction data from 59 Grade 8 students. It also analyzes performance differences across the Triangles, Solids, and Quadrilaterals modules to identify topic-specific learning challenges.

II. Literature Review

Educational technologies have increasingly been used to support learning in science, technology, engineering, and mathematics (STEM) education. Among these technologies, Augmented Reality (AR) has received significant attention because it allows learners to interact with digital content within a real-world environment. AR is particularly useful for subjects that require visualization, manipulation, and spatial reasoning. Geometry is one such area, as many concepts depend on students' ability to mentally construct, transform, and interpret shapes and spatial relationships [1], [6].

Traditional geometry instruction often relies on textbook diagrams, board drawings, and static two-dimensional representations. While these methods can introduce basic definitions and properties, they may not fully support learners who struggle to visualize three-dimensional objects. Such limitations can reduce students' ability to understand geometric relationships and may affect their engagement with abstract mathematical concepts [2]. Previous studies have shown that interactive digital tools can improve learners' ability to understand geometric relationships by allowing direct manipulation of visual objects [3], [5]. AR has been explored as a solution to these limitations because it can represent geometric figures as interactive three-dimensional models. Students can rotate, zoom, and observe objects from multiple angles, which may support the development of spatial reasoning and conceptual understanding. Studies on AR-based mathematics learning have reported improvements in engagement, visual thinking, academic motivation, and problem-solving ability [6], [9], [10]. In particular, AR-based learning environments have shown potential in helping students visualize geometric shapes more effectively than traditional flat illustrations [6].

Several AR-based tools have been developed for mathematics and STEM education. Some studies have focused on the use of AR to improve mathematical comprehension, while others have explored AR in relation to visual thinking, academic motivation, and problem-solving. For example, Bankar et al. [8] highlighted the role of immersive technologies such as AR and VR in improving students' understanding of abstract learning content. Similarly, Elsayed and Al-Najrani [9] reported that AR improved visual thinking abilities and academic motivation among middle school mathematics learners. Research by Nindiasari et al. [12] further showed that AR-based geometry learning, when integrated with a STEAM-based approach, can support students' problem-solving skills. Despite these benefits, the effectiveness of AR depends not only on the technology itself but also on instructional design, curriculum alignment, usability, and the quality of learning activities. Many existing AR solutions are designed for broad educational contexts and may not directly address the specific needs of Sri Lankan classrooms. In addition, several AR-based educational tools require high-end devices, continuous internet connectivity, or frequent software updates, which can limit their use in resource-constrained learning environments [5], [6]. These limitations are particularly important in contexts where access to advanced digital infrastructure is uneven.

A key limitation in resource-constrained educational environments is the dependence of many digital tools on high-end hardware and stable internet connectivity. In many Sri Lankan schools, such resources may not be consistently available. Therefore, AR applications intended for local classroom use should be lightweight, offline-capable, and aligned with the national curriculum. Previous work on AR application development using Unity and Vuforia demonstrates that marker-based AR can provide a practical development approach for interactive educational applications [7]. However, there remains a need for localized AR systems that are designed specifically for Grade 8 geometry learning in Sri Lanka.

Several authors have also emphasized the importance of localizing educational technologies. Yanuarto [4] noted that context-sensitive AR tools can improve students' comprehension of mathematical concepts. Similarly, curriculum-specific AR tools have been identified as useful for bridging the gap between theoretical learning and practical interaction [19]. These findings suggest that AR applications should not only be technologically functional but also pedagogically relevant and adaptable to learners' local educational needs.

The reviewed literature indicates that AR has strong potential to support geometry learning through visualization and interaction. However, there remains a gap in offline-capable, curriculum-aligned AR tools designed for Grade 8 geometry education in Sri Lanka. This study addresses that gap by developing and evaluating a mobile AR application that supports interactive learning under local technical and educational constraints.

III. Methodology

This study used a quantitative research design to evaluate the effectiveness of an Augmented Reality (AR) application for Grade 8 geometry learning. The evaluation focused on two main dimensions: student engagement and quiz-based academic performance. Student interaction data were automatically logged by the application and analyzed to identify patterns in AR usage, topic-wise performance, and the relationship between interaction time and quiz scores.

A. System Architecture

The AR application was developed using the Unity 3D game engine and the Vuforia Software Development Kit (SDK). Unity was used to create the interactive learning environment, while Vuforia supported marker-based AR tracking. The application was designed to operate on low-specification Android or iOS mobile devices with at least 2 GB of RAM and standard camera functionality. This design decision was made to improve accessibility in resource-constrained classrooms.

The system architecture consists of three integrated components: AR content management, the user interaction interface, and the data logging and analysis module. The AR content

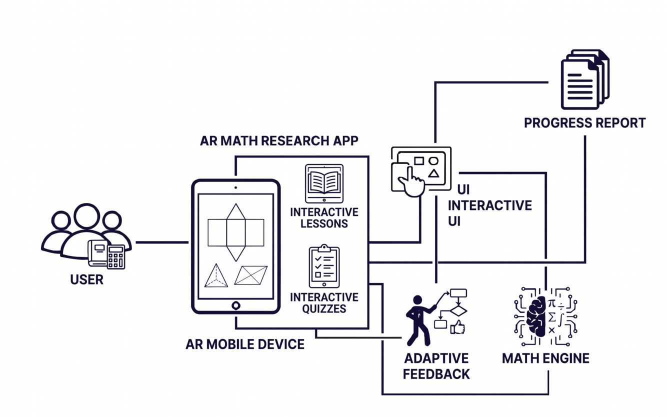

Fig. 1. System architecture of the proposed AR geometry learning application

management component loads 3D geometry models when a student scans a printed worksheet containing Vuforia image targets. Each marker is linked to a specific geometry topic, such as Triangles, Solids, or Quadrilaterals. When the marker is detected, the corresponding 3D model is displayed on the mobile device screen and aligned with the real-world worksheet. The user interaction interface enables students to manipulate the displayed 3D models using touch gestures such as rotation, zooming, and panning. It also provides access to quiz activities related to the displayed geometry content, encouraging active exploration rather than passive viewing. The data logging and analysis module records student interactions automatically during each session, including total session time, AR interaction time, quiz time, rotation count, button clicks, answer clicks, and quiz scores. These data are stored in two main files: the Session Summary File and the Quiz Performance File. The Session Summary File records overall usage and engagement information, while the Quiz Performance File records quiz-specific performance data.

Fig. 1 presents the overall system architecture of the proposed AR application. The architecture shows how the learner interacts with the mobile AR interface, how geometry models and quiz activities are delivered, and how interaction data are recorded for later analysis.

B. System Workflow

The learning process begins when a student opens the AR application and scans a worksheet containing a marker. Once the Vuforia SDK detects the marker, the application displays the relevant 3D geometry model. Students can then interact with the model by rotating, zooming, and viewing it from different angles. After interacting with the model, students complete quiz questions related to the geometry concept. The quiz module awards points for correct answers and provides feedback for incorrect responses. The total quiz score is recorded together with the time taken to complete the quiz. Throughout the session, all interaction data are logged automatically without interrupting the learning process.

In the complete workflow, the student first scans the AR marker on the worksheet, after which the application detects the marker through Vuforia and displays the corresponding 3D geometry model. The student then interacts with the model using touch-based gestures and proceeds to answer quiz questions related to the displayed geometry content. During this process, engagement and quiz performance data are recorded by the application and later analyzed to evaluate learning outcomes.

Fig. 2. User interaction interface of the AR geometry learning application

Fig. 2 shows the user interaction interface used by students during the learning activity. The interface enables students to view and manipulate geometry models through touch-based interactions and to access quiz questions related to the displayed content.

C. Participants and Data Collection

The study was conducted with 59 Grade 8 students. Each student used the AR application during a geometry learning session, and the application recorded interaction data at the session level. The collected data included total session duration, AR interaction time, quiz completion time, rotation counts, button clicks, answer clicks, and quiz scores. The quiz scores were measured on a 0–5 scale for each relevant geometry topic. The three main lesson sections analyzed in the study were Triangles, Solids, and Quadrilaterals, as these topics are important components of Grade 8 geometry and require visual understanding of shapes and spatial relationships. To compare learning outcomes, students completed assessments under traditional textbook-based learning and AR-supported learning conditions. The traditional condition was used as the baseline, while the AR-supported condition measured performance after students interacted with the Shilpa AR application. The comparison was used to examine whether AR-supported learning showed improvement over the conventional approach.

D. Data Analysis

The collected data were processed using Python, with the pandas library used for statistical analysis and data handling. The analysis included descriptive statistics, comparative analysis, correlation analysis, and visual analysis. Descriptive statistics, including means, medians, standard deviations, and totals, were calculated for key variables such as session time, AR interaction time, quiz time, quiz scores, rotation counts, button clicks, and answer clicks. Comparative analysis was used to examine student performance across the Triangles, Solids, and Quadrilaterals modules, allowing topic-wise differences in learning outcomes to be identified. Correlation analysis was used to examine the relationship between AR interaction time and quiz performance, with the purpose of determining whether increased time spent interacting with AR content was associated with higher quiz scores. Visual analysis was also conducted using graphs such as histograms, bar charts, boxplots, and scatter plots to examine engagement patterns, score distributions, and differences between learning modules.

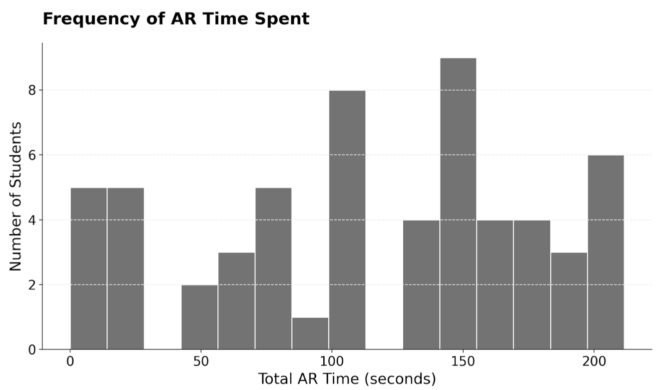

Fig. 3. Frequency distribution of AR interaction time

Fig. 3 illustrates the distribution of AR interaction time among students. This visualization was used to examine how long students interacted with the AR models during the learning sessions.

E. Ethical Considerations

The study followed ethical practices for educational research involving school students. The study was conducted with school-level permission, and participant data were anonymized before analysis. Participants were assigned sequential student codes from S001 to S059, and no personally identifiable information was included in the dataset. The data were analyzed only at the session level to protect student privacy. Informed consent was obtained before data collection, and the study was conducted in accordance with ethical guidelines for educational technology research.

IV. Results and Discussion

This section presents the results of the study and discusses the effectiveness of the AR application in supporting Grade 8 geometry learning. The analysis focuses on student engagement, quiz performance, topic-wise differences, and the relationship between AR interaction time and academic outcomes.

The first aspect analyzed was student engagement with the AR application. The total session time showed that students spent between 350 and 600 seconds using the platform. This indicates that students were able to remain engaged long enough to explore the 3D geometry models and complete the associated quiz activities. The interaction logs also showed that students used touch-based gestures such as rotation and model manipulation, suggesting that the AR application encouraged active participation rather than passive viewing. As shown in Fig. 3, most students spent between 100 and 150 seconds interacting directly with the AR content. However, the variation in AR interaction time suggests that students used the application at different levels of intensity. Some students may have explored the 3D models more actively, while others may have moved more quickly to the quiz component.

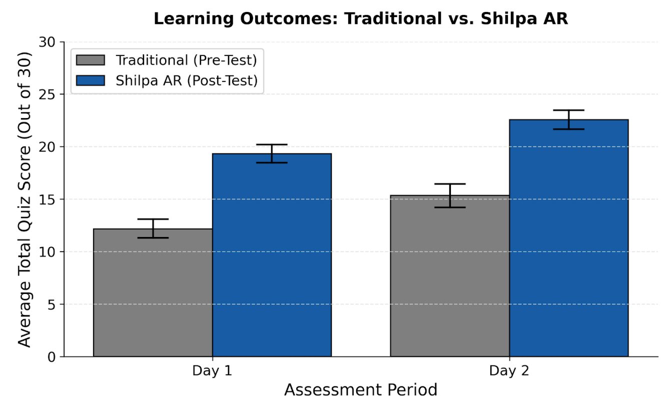

Fig. 4. Learning outcomes: traditional learning vs. Shilpa AR

The engagement metrics indicate that the application successfully encouraged students to interact with geometry content in a more active way than conventional textbook-based learning. However, engagement time alone does not fully represent learning quality. The nature of the interaction, the clarity of the learning task, and the instructional support provided during the activity are also important factors that influence learning outcomes.

The learning outcome comparison showed improvement in the AR-supported learning condition when compared with the traditional learning condition represented in the assessment results. Fig. 5 presents the score distribution comparing traditional textbook learning with Shilpa AR-supported learning. The AR-supported scores were higher, suggesting that the use of interactive 3D models may have helped students understand selected geometry concepts more effectively. Nevertheless, this improvement should be interpreted carefully because the study also found that increased AR interaction time did not strongly correspond with higher quiz performance. Therefore, the improvement cannot be explained by interaction time alone. Other factors, such as the clarity of the 3D models, quiz design, prior knowledge, teacher support, and topic difficulty, may also have influenced the results.

Student performance also differed across the three geometry topics: Triangles, Solids, and Quadrilaterals. The quiz scores were recorded on a 0–5 scale for each topic. The Solids module recorded the highest mean score of 2.81, followed

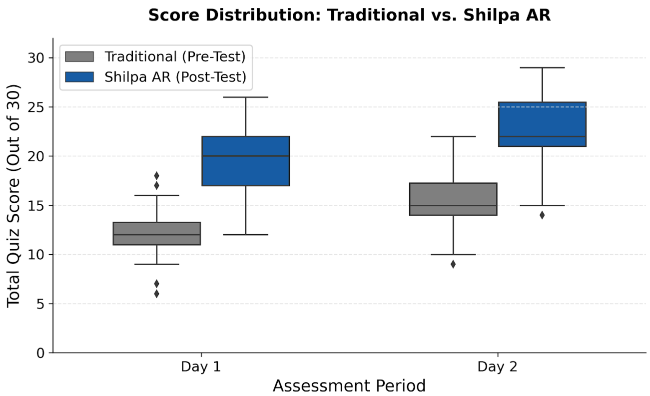

Fig. 5. Score distribution: traditional learning vs. Shilpa AR

by the Quadrilaterals module with a mean score of 2.65. In contrast, the Triangles module recorded the lowest mean score of 2.10. The overall mean quiz score was 2.51 out of 5. These results indicate that students performed better in the Solids and Quadrilaterals modules than in the Triangles module. One possible explanation is that solid shapes may benefit more directly from 3D visualization because learners can observe faces, edges, and vertices from multiple angles. Quadrilaterals may also be easier to represent through interactive visual models. In contrast, triangle-related concepts may require additional explanation, guided practice, or more carefully designed interactive activities.

Fig. 6 presents the average user engagement metrics across the learning activities. The engagement patterns suggest that students interacted actively with the AR content, but the differences in quiz performance across topics show that AR was not equally effective for all geometry concepts. This indicates that some topics may require additional instructional scaffolding, such as step-by-step explanations, guided hints, or teacher-led discussion after AR interaction.

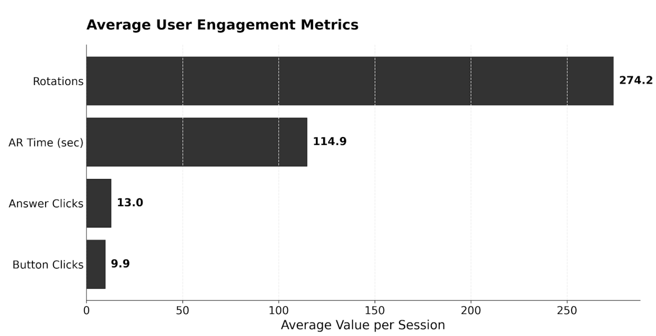

Fig. 6. Average user engagement metrics

The relationship between AR interaction time and quiz performance was examined to determine whether students who spent more time interacting with the AR models achieved higher quiz scores. The results showed only a weak relationship between AR interaction time and quiz performance, indicating that spending more time with the AR content did not necessarily lead to better academic performance. This finding is important because it shows that AR engagement should not be treated as equivalent to learning achievement. A student may spend more time rotating or viewing a model without necessarily developing a deeper conceptual understanding.

Fig. 7 shows the distribution of 3D model rotations. The variation in rotation counts further highlights differences in how students interacted with the models. Although rotation and model manipulation indicate active engagement, these actions did not strongly correspond with improved quiz performance. This suggests that the effectiveness of AR depends not only on the number of interactions but also on the quality and purpose of those interactions.

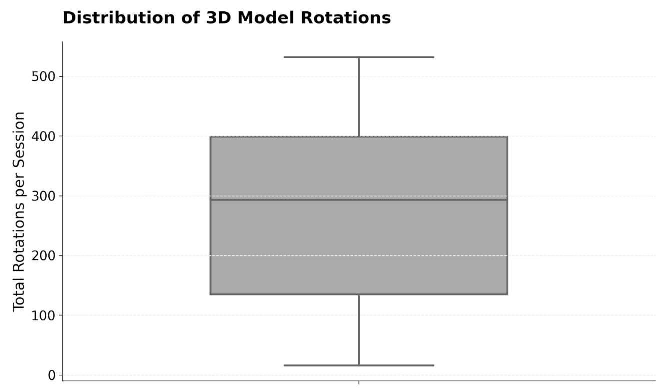

Fig. 7. Distribution of 3D model rotations

The score distribution also showed performance variation among students. Some students achieved relatively higher quiz scores, while others continued to struggle despite interacting with the AR application. This variation may be influenced by differences in prior knowledge, learning pace, digital familiarity, motivation, and the level of support provided during the activity. Students with stronger spatial reasoning skills may have benefited more quickly from the AR models, whereas students with weaker foundational knowledge may have required additional guidance.

Overall, the findings indicate that the AR application successfully supported student engagement in geometry learning. By allowing students to manipulate 3D models, the application provided a more interactive learning experience than static textbook illustrations. This is particularly relevant for geometry topics that require visualization of spatial structures. However, the weak relationship between interaction time and quiz performance shows that technology alone is not sufficient to ensure learning improvement. The educational effectiveness of AR depends on how well the technology is integrated with instructional design. Students need guidance to understand what they should observe, compare, and conclude while interacting with 3D models. The lower performance in the Triangles module suggests that some geometry topics may require more structured explanations than others. While AR can help visualize objects, it may not automatically explain mathematical reasoning, relationships, or problem-solving steps. Therefore, AR should be used as a supportive learning tool rather than a replacement for instruction. Future versions of the application could include guided prompts, concept explanations, adaptive feedback, and structured tasks that direct students toward important geometric properties. Such features may help convert interaction into meaningful learning.

The results also highlight the importance of designing AR learning systems for local classroom conditions. Since the application was designed to work offline and on low-specification devices, it addresses practical limitations found in many Sri Lankan schools. This makes the proposed system more suitable for resource-constrained learning environments. However, the findings also show that AR should be combined with guided instruction, feedback, and topic-specific learning support to improve academic performance more effectively.

V. Conclusion

This study developed and evaluated an offline-capable Augmented Reality (AR) application for Grade 8 geometry learning in Sri Lanka. The application transformed selected geometry concepts into interactive three-dimensional models and included quiz activities to support student learning. A quantitative evaluation conducted with 59 Grade 8 students showed that the application successfully encouraged active engagement, with students spending between 350 and 600 seconds using the platform.

The results showed that students achieved an overall mean quiz score of 2.51 out of 5. Performance was higher in the Solids and Quadrilaterals modules than in the Triangles module, suggesting that some geometry topics may benefit more directly from AR-based visualization than others. However, the study did not find a strong relationship between AR interaction time and quiz performance. This indicates that while AR can increase student engagement, engagement alone does not guarantee improved academic achievement. The findings suggest that AR has potential as a supportive tool for geometry education in Sri Lanka, especially in classrooms where access to advanced digital infrastructure is limited. At the same time, the results highlight the need to combine AR-based visualization with structured instructional support. Future improvements should focus on adding guided explanations, adaptive feedback, personalized learning paths, gamified activities, and collaborative learning features. Further studies using controlled experimental designs and longitudinal evaluations are also recommended to examine the long-term impact of AR-supported geometry learning.

In conclusion, the proposed AR application provides an accessible and engaging approach to Grade 8 geometry learning. Its educational impact can be strengthened by integrating interactive visualization with classroom-based guidance, topic-specific scaffolding, and improved instructional design.

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